

Readable Version of `commutation-sl3.nb`

This document is a readable transcription of the executable Mathematica notebook cells from `commutation-sl3.nb`. Notebook metadata and timestamp fields were omitted; inputs and outputs are preserved in order. Matrix definitions are typeset in mathematical form for readability.

Input cell 1

```
[blank input cell]
```

Input cell 2

```
(* U(SL(3)) *)
```

Input cell 3: matrix definitions and initialization

MatH12

$$\text{MatH12} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

MatH23

$$\text{MatH23} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

MatH13

$$\text{MatH13} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

MatE13

$$\text{MatE13} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

MatE12

$$\text{MatE12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

MatE23

$$\text{MatE23} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

MatE31

$$\text{MatE31} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

MatE32

$$\text{MatE32} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

MatE21

$$\text{MatE21} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

```
MatSL3 = {MatH12, MatH23, MatH13, MatE12, MatE13, MatE23,  
          MatE21, MatE31, MatE32};  
  
Clear[com, p1, p2, i, j, q, q1, ord]
```

Input cell 4: symbolic generators and automatic commutator table construction

```
sl3 = {H12, H23, H13, E12, E13, E23, E21, E31, E32};  
  
ord = Table[0, {i, 1, 3}, {j, 1, 3}];  
ord[[1, 2]] = 4; ord[[1, 3]] = 5; ord[[2, 3]] = 6;  
ord[[2, 1]] = 7; ord[[3, 1]] = 8; ord[[3, 2]] = 9;  
  
For[i = 1, i < 10, i++,  
  For[j = 1, j < 10, j++,  
  
    mat = MatSL3[[i]].MatSL3[[j]] - MatSL3[[j]].MatSL3[[i]];  
  
    If[Norm[mat] == 0,  
      com[sl3[[i]], sl3[[j]]] = 0,  
      If[Tr[Abs[mat]] != 0,  
        q = 1;  
        While[mat[[q, q]] != 0, q++];  
        q1 = Mod[q, 3] + 1;  
        com[sl3[[i]], sl3[[j]]] =  
          (mat[[q1, q1]] / MatSL3[[q1, q1, q1]]) sl3[[q1]],  
        p1 = 1;  
        p2 = 1;  
        While[mat[[p1, p2]] == 0,  
          p2++;  
          If[Mod[p2, 3] == 1, p1++; p2 = 1]  
        ];  
        com[sl3[[i]], sl3[[j]]] =  
          (mat[[p1, p2]] / MatSL3[[ord[[p1, p2]], p1, p2]])  
          sl3[[ord[[p1, p2]]]]  
      ]  
    ]  
  ]  
]
```

Input cell 5

```
[blank input cell]
```

Input/Output cells: selected commutation relations

$$\begin{aligned}\text{com}(H12, E12) &= 2E12, \\ \text{com}(H12, E13) &= E13, \\ \text{com}(H12, E23) &= -E23, \\ \text{com}(H12, E21) &= -2E21, \\ \text{com}(H12, E31) &= -E31, \\ \text{com}(H12, E32) &= E32, \\ \text{com}(H12, H13) &= 0, \\ \text{com}(H12, H23) &= 0, \\ \text{com}(E12, E21) &= H12.\end{aligned}$$

Same relations in notebook input/output order

```
In:  com[H12, E12]
Out: 2 E12

In:  com[H12, E13]
Out: E13

In:  com[H12, E23]
Out: -E23

In:  com[H12, E21]
Out: -2 E21

In:  com[H12, E31]
Out: -E31

In:  com[H12, E32]
Out: E32

In:  com[H12, H13]
Out: 0

In:  com[H12, H23]
Out: 0

In:  com[E12, E21]
Out: H12
```

Appendix: Complete Rewriting Relations Used by the C++ Program

The Mathematica notebook constructs commutators from the standard matrix representation of $sl(3)$. The C++ implementation uses the following product rewriting rules to normalize expressions into the selected canonical ordering:

E31, E21, E32, H12, H23, E23, E12, E13

The symbol H13 is expanded and is not used as an independent basis generator.

$H13 = H12 + H23$	$E23 * H12 = H12 * E23 + E23$
$E13 * E12 = E12 * E13$	$E23 * E32 = E32 * E23 + H12 + H23$
$E13 * E23 = E23 * E13$	$E23 * E21 = E21 * E23$
$E13 * H23 = H23 * E13 - 2E13$	$E23 * E31 = E31 * E23 + E21$
$E13 * H12 = H12 * E13 - E13$	$H23 * H12 = H12 * H23$
$E13 * E32 = E32 * E13 + E12$	$H23 * E32 = E32 * H23 - E32$
$E13 * E21 = E21 * E13 - E23$	$H23 * E21 = E21 * H23 - E21$
$E13 * E31 = E31 * E13 + H23$	$H23 * E31 = E31 * H23 - 2E31$
$E12 * E23 = E23 * E12 + E13$	$H12 * E32 = E32 * H12 + E32$
$E12 * H23 = H23 * E12 - E12$	$H12 * E21 = E21 * H12 - 2E21$
$E12 * H12 = H12 * E12 - 2E12$	$H12 * E31 = E31 * H12 - E31$
$E12 * E32 = E32 * E12$	$E32 * E21 = E21 * E32 + E31$
$E12 * E21 = E21 * E12 + H12$	$E32 * E31 = E31 * E32$
$E12 * E31 = E31 * E12 - E32$	$E21 * E31 = E31 * E21$
$E23 * H23 = H23 * E23 - E23$	

These rules are the ones encoded in `polynomial::transform()`. Changing `convert()`, `unconvert()` and `polynomial::transform()` changes the generator ordering and the algebra-specific rewriting rules.